The Journal of Open Source Software

# tensorflow-riemopt: a library for optimization on Riemannian manifolds in TensorFlow

**Oleg Smirnov**[1¶]

**1** Microsoft Gaming, Stockholm, Sweden ¶ Corresponding author

**DOI:** 10.xxxxxx/draft

**Software**

- Review
- Repository
- Archive

**Editor:** Neea Rusch

**Reviewers:**

- @Fastaxx
- @yossibokorbleile
- @zazabap

**Submitted:** 30 October 2026
**Published:** unpublished

**License**
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

## Summary

tensorflow-riemopt is a TensorFlow (Abadi et al., 2016) library for training machine learning models whose parameters are constrained to Riemannian manifolds. It provides multiple manifolds with their Riemannian metrics, exponential and logarithmic maps, geodesics, retractions and vector transports, implemented either as closed-form expressions or approximations where applicable. The library also implements Riemannian versions of stochastic gradient descent (Bonnabel, 2013) and of the adaptive RMSprop (Roy et al., 2018) and Adam (Bécigneul & Ganea, 2019) algorithms. After a manifold is attached to a TensorFlow variable, Keras layers and training loops can be used without modification, resulting in a geometry-aware model that can be trained and served inside a TensorFlow Extended pipeline (Baylor et al., 2017).

The repository includes example implementations of SPDNet (Huang & Van Gool, 2017), GrNet (Huang et al., 2018), and LieNet (Huang et al., 2017) methods built on the library.

## Statement of need

Several families of machine learning models keep their parameters on a non-Euclidean space. Hyperbolic embeddings represent data with latent hierarchies and are used in language modeling (Ganea et al., 2018; Nickel & Kiela, 2017, 2018; Sala et al., 2018), recommendation (Chamberlain et al., 2019) and image classification (Khrulkov et al., 2020). Weights on the Stiefel and Grassmann manifolds appear in networks for emotion and action recognition from video (Chakraborty & Vemuri, 2019; Huang et al., 2018; Huang & Van Gool, 2017) and in image set classification (Y. Li et al., 2015). Symmetric positive definite matrices represent diffusion tensors and covariance descriptors (Pennec et al., 2006), and the Lie groups $\mathrm{SO}(3)$ and $\mathrm{SE}(3)$ parameterize pose estimation and skeleton-based action recognition (Hou et al., 2018; Huang et al., 2017). Training of these models requires an optimizer whose updates are constrained to the manifold, which is not guaranteed by the conventional Euclidean optimizers.

## State of the field

Riemannian stochastic optimization (Bécigneul & Ganea, 2019; Bonnabel, 2013; Kasai et al., 2019; Roy et al., 2018) has support in PyTorch through Geoopt (Kochurov et al., 2020) and McTorch (Meghwanshi et al., 2018). TensorFlow, and the related TFX and TensorFlow Serving frameworks, had no maintained equivalent. tensorflow-riemopt closes this gap by providing manifold-constrained variables and Riemannian optimizers that replace their Euclidean counterparts in any existing TensorFlow / Keras pipeline. It also exposes a low-level API for the geometric operations, so new manifolds and algorithms are implemented and tested against the existing alternatives.

DRAFT

Across related work, Geoopt (Kochurov et al., 2020) and McTorch (Meghwanshi et al., 2018) implement Riemannian stochastic optimizers for PyTorch, while Pymanopt (Townsend et al., 2016) implements deterministic solvers and differentiates cost functions supported by TensorFlow, PyTorch, and JAX backends. However, Pymanopt computes the manifold operations in NumPy and does not provide stochastic optimizers, so it does not support minibatch training of neural networks. Geomstats (Miolane et al., 2020) implements additional manifolds and differential-geometric constructions, providing a scikit-learn API (Pedregosa et al., 2011) with NumPy, PyTorch, and Autograd backends. Manopt (Boumal et al., 2014), Manifolds.jl (Axen et al., 2023) and Rieoptax (Utpala et al., 2022) provide alternative implementations in MATLAB, Julia, and JAX correspondingly.

## Software design

In tensorflow-riemopt, the manifold constraint is a property of the variable, that can be set by `assign_to_manifold(variable, manifold)` function. Therefore, a model can combine manifold-constrained and unconstrained layers, which are treated accordingly by the Riemannian optimizers. After training, the exported model contains only standard TensorFlow operations, so TensorFlow TFX components do not depend on the library.

```
import tensorflow as tf
from tensorflow_riemopt.variable import assign_to_manifold
from tensorflow_riemopt.manifolds import StiefelEuclidean
from tensorflow_riemopt.optimizers import RiemannianSGD


class BiMap(tf.keras.layers.Layer):
    """Bilinear mapping layer of SPDNet with a weight on the Stiefel manifold."""

    def __init__(self, output_dim, **kwargs):
        super().__init__(**kwargs)
        self.output_dim = output_dim

    def build(self, input_shape):
        self.w = self.add_weight(name="w", shape=[input_shape[-1], self.output_dim])
        assign_to_manifold(self.w, StiefelEuclidean())

    def call(self, inputs):
        return tf.transpose(self.w) @ inputs @ self.w


model = tf.keras.Sequential([BiMap(32), tf.keras.layers.Flatten(), tf.keras.layers.Dense
model.compile(optimizer=RiemannianSGD(learning_rate=0.01), loss="sparse_categorical_cros
```

As in Manopt (Boumal et al., 2014), each manifold subclasses `Manifold` and implements the methods the optimization algorithms call: the Riemannian metric `inner(x, u, v)`, the tangent-space projection `proju(x, u)`, a retraction `retr(x, u)` and a vector transport `transp(x, y, v)`. Exponential and logarithmic maps and parallel transport are additional methods that can be added for adaptive optimization support.

```
from tensorflow_riemopt.manifolds import Sphere

S = Sphere()
x = S.projx(tf.constant([0.1, -0.1, 0.1]))
u = S.proju(x, tf.constant([1.0, 1.0, 1.0]))
v = S.proju(x, tf.constant([-0.7, -1.4, 1.4]))
y = S.exp(x, v)
```

DRAFT

```
u_y = S.transp(x, y, u)
v_y = S.transp(x, y, v)
```

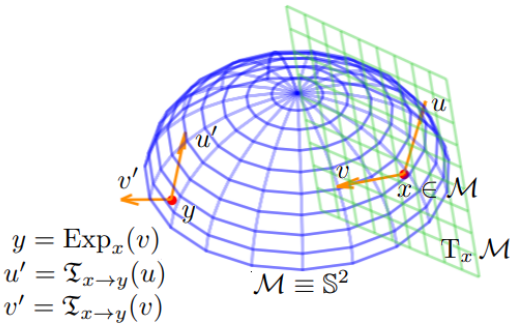



**Figure 1:** Geometric operations on the sphere $\mathbb{S}^2$. The exponential map takes $x$ along $v$ to $y$, and vector transport carries $u$ and $v$ from $T_x\mathcal{M}$ to $T_y\mathcal{M}$.

The library implements the exponential and logarithmic map in closed form where it is known. Otherwise, it falls back to a published approximate algorithm, e.g., the matrix-algebraic logarithm on the Stiefel manifold under the canonical metric (Zimmermann, 2017) and the Cayley retraction (J. Li et al., 2020). Every manifold method is composed of TensorFlow operations, so automatic differentiation, graph execution and eager execution do not require manifold-specific code.

The optimizers implement the Keras optimizer interface, including the sparse update path for `tf.IndexedSlices`, so in an embedding table on the Poincaré ball or the hyperboloid only the rows present in a batch are updated. The `layers` module provides such a table as `ManifoldEmbedding`. Optimizer state and hyperparameters are serialized through the Keras configuration mechanism, so a compiled model is saved and restored with the standard calls.

## Research impact statement

As of 30 October 2026, Google Scholar lists 12 works citing the preprint (Smirnov, 2021), the GitHub repository has ~100 stars and ~20 forks. ANTHEM (Choudhary et al., 2022), a hyperbolic entity model for product search, imports the library in the public implementation released by Amazon Science, which installs it from our repository under its former name tensorflow-manopt. The monograph on Riemannian geometry and geometric statistics by Guigui et al. (2023) and the survey of geometric optimization for deep learning by Fei et al. (2025) cite the library, and Axen et al. (2023) include it in their cross-library benchmark of manifold operations. The citing research includes Riemannian Hamiltonian methods for min-max optimization (Han et al., 2023), drug-target interaction prediction (Binatlı & Gönen, 2023) and differentially private Fréchet means on symmetric positive definite matrices (Utpala et al., 2023).

The package is distributed on PyPI, documented on Read the Docs, and tested in continuous integration with coverage reporting against current TensorFlow releases.

## AI usage disclosure

A generative AI tool was used to prepare the manuscript draft from an earlier preprint describing the library (Smirnov, 2021). The final draft was reviewed and edited to verify its claims against the repository and the cited works.

DRAFT

The Journal of Open Source Software

## Acknowledgements

We thank the users who reported issues and contributed fixes to the repository.

## References

Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., Devin, M., Ghemawat, S., Irving, G., Isard, M., & others. (2016). TensorFlow: A system for large-scale machine learning. *12th USENIX Symposium on Operating Systems Design and Implementation*, 265–283.

Axen, S. D., Baran, M., Bergmann, R., & Rzecki, K. (2023). Manifolds.jl: An extensible Julia framework for data analysis on manifolds. *ACM Transactions on Mathematical Software*, *49*(4), 1–23. https://doi.org/10.1145/3618296

Baylor, D., Breck, E., Cheng, H.-T., Fiedel, N., Foo, C. Y., Haque, Z., Haykal, S., Ispir, M., Jain, V., Koc, L., & others. (2017). TFX: A TensorFlow-based production-scale machine learning platform. *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1387–1395. https://doi.org/10.1145/3097983.3098021

Bécigneul, G., & Ganea, O.-E. (2019). Riemannian adaptive optimization methods. *International Conference on Learning Representations*.

Binatlı, O. C., & Gönen, M. (2023). MOKPE: Drug–target interaction prediction via manifold optimization based kernel preserving embedding. *BMC Bioinformatics*, *24*, 276. https://doi.org/10.1186/s12859-023-05401-1

Bonnabel, S. (2013). Stochastic gradient descent on Riemannian manifolds. *IEEE Transactions on Automatic Control*, *58*(9), 2217–2229. https://doi.org/10.1109/TAC.2013.2254619

Boumal, N., Mishra, B., Absil, P.-A., & Sepulchre, R. (2014). Manopt, a Matlab toolbox for optimization on manifolds. *Journal of Machine Learning Research*, *15*(42), 1455–1459.

Chakraborty, R., & Vemuri, B. C. (2019). Statistics on the Stiefel manifold: Theory and applications. *The Annals of Statistics*, *47*(1), 415–438. https://doi.org/10.1214/18-AOS1692

Chamberlain, B. P., Hardwick, S. R., Wardrope, D. R., Dzogang, F., Daolio, F., & Vargas, S. (2019). Scalable hyperbolic recommender systems. *arXiv Preprint arXiv:1902.08648*. https://doi.org/10.48550/arXiv.1902.08648

Choudhary, N., Rao, N., Katariya, S., Subbian, K., & Reddy, C. K. (2022). ANTHEM: Attentive hyperbolic entity model for product search. *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*, 161–171. https://doi.org/10.1145/3488560.3498456

Fei, Y., Liu, Y., Jia, C., Li, Z., Wei, X., & Chen, M. (2025). A survey of geometric optimization for deep learning: From Euclidean space to Riemannian manifold. *ACM Computing Surveys*, *57*(5), 1–37. https://doi.org/10.1145/3708498

Ganea, O.-E., Bécigneul, G., & Hofmann, T. (2018). Hyperbolic neural networks. *Advances in Neural Information Processing Systems*, *31*, 5350–5360.

Guigui, N., Miolane, N., & Pennec, X. (2023). Introduction to Riemannian geometry and geometric statistics: From basic theory to implementation with Geomstats. *Foundations and Trends in Machine Learning*, *16*(3), 329–493. https://doi.org/10.1561/2200000098

Han, A., Mishra, B., Jawanpuria, P., Kumar, P., & Gao, J. (2023). Riemannian Hamiltonian methods for min-max optimization on manifolds. *SIAM Journal on Optimization*, *33*(3), 1797–1827. https://doi.org/10.1137/22M1492684

DRAFT

Hou, B., Miolane, N., Khanal, B., Lee, M. C. H., Alansary, A., McDonagh, S., Hajnal, J. V., Rueckert, D., Glocker, B., & Kainz, B. (2018). Computing CNN loss and gradients for pose estimation with Riemannian geometry. *Medical Image Computing and Computer Assisted Intervention*, 756–764. https://doi.org/10.1007/978-3-030-00928-1_85

Huang, Z., & Van Gool, L. (2017). A Riemannian network for SPD matrix learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, *31*. https://doi.org/10.1609/aaai.v31i1.10866

Huang, Z., Wan, C., Probst, T., & Van Gool, L. (2017). Deep learning on Lie groups for skeleton-based action recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 6099–6108. https://doi.org/10.1109/CVPR.2017.137

Huang, Z., Wu, J., & Van Gool, L. (2018). Building deep networks on Grassmann manifolds. *Proceedings of the AAAI Conference on Artificial Intelligence*, *32*. https://doi.org/10.1609/aaai.v32i1.11725

Kasai, H., Jawanpuria, P., & Mishra, B. (2019). Riemannian adaptive stochastic gradient algorithms on matrix manifolds. *Proceedings of the 36th International Conference on Machine Learning*, 3262–3271.

Khrulkov, V., Mirvakhabova, L., Ustinova, E., Oseledets, I., & Lempitsky, V. (2020). Hyperbolic image embeddings. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 6418–6428. https://doi.org/10.1109/CVPR42600.2020.00645

Kochurov, M., Karimov, R., & Kozlukov, S. (2020). Geoopt: Riemannian optimization in PyTorch. *arXiv Preprint arXiv:2005.02819*. https://doi.org/10.48550/arXiv.2005.02819

Li, J., Li, F., & Todorovic, S. (2020). Efficient Riemannian optimization on the Stiefel manifold via the Cayley transform. *International Conference on Learning Representations*.

Li, Y., Wang, R., Huang, Z., Shan, S., & Chen, X. (2015). Face video retrieval with image query via hashing across Euclidean space and Riemannian manifold. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4758–4767. https://doi.org/10.1109/CVPR.2015.7299108

Meghwanshi, M., Jawanpuria, P., Kunchukuttan, A., Kasai, H., & Mishra, B. (2018). McTorch, a manifold optimization library for deep learning. *arXiv Preprint arXiv:1810.01811*. https://doi.org/10.48550/arXiv.1810.01811

Miolane, N., Guigui, N., Le Brigant, A., Mathe, J., Hou, B., Thanwerdas, Y., Heyder, S., Peltre, O., Koep, N., Zaatiti, H., & others. (2020). Geomstats: A Python package for Riemannian geometry in machine learning. *Journal of Machine Learning Research*, *21*(223), 1–9.

Nickel, M., & Kiela, D. (2018). Learning continuous hierarchies in the Lorentz model of hyperbolic geometry. *Proceedings of the 35th International Conference on Machine Learning*, 3779–3788.

Nickel, M., & Kiela, D. (2017). Poincaré embeddings for learning hierarchical representations. *Advances in Neural Information Processing Systems*, *30*, 6338–6347.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., & others. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Pennec, X., Fillard, P., & Ayache, N. (2006). A Riemannian framework for tensor computing. *International Journal of Computer Vision*, *66*(1), 41–66. https://doi.org/10.1007/s11263-005-3222-z

Roy, S. K., Mhammedi, Z., & Harandi, M. (2018). Geometry aware constrained optimization techniques for deep learning. *Proceedings of the IEEE Conference on Computer Vision and*

DRAFT

*Pattern Recognition*, 4460–4469. https://doi.org/10.1109/CVPR.2018.00469

Sala, F., De Sa, C., Gu, A., & Ré, C. (2018). Representation tradeoffs for hyperbolic embeddings. *Proceedings of the 35th International Conference on Machine Learning*, 4460–4469.

Smirnov, O. (2021). TensorFlow RiemOpt: A library for optimization on Riemannian manifolds. *arXiv Preprint arXiv:2105.13921*. https://doi.org/10.48550/arXiv.2105.13921

Townsend, J., Koep, N., & Weichwald, S. (2016). Pymanopt: A Python toolbox for optimization on manifolds using automatic differentiation. *Journal of Machine Learning Research*, *17*(137), 1–5.

Utpala, S., Han, A., Jawanpuria, P., & Mishra, B. (2022). Rieoptax: Riemannian optimization in JAX. *arXiv Preprint arXiv:2210.04840*. https://doi.org/10.48550/arXiv.2210.04840

Utpala, S., Vepakomma, P., & Miolane, N. (2023). Differentially private Fréchet mean on the manifold of symmetric positive definite (SPD) matrices with log-Euclidean metric. *Transactions on Machine Learning Research*.

Zimmermann, R. (2017). A matrix-algebraic algorithm for the Riemannian logarithm on the Stiefel manifold under the canonical metric. *SIAM Journal on Matrix Analysis and Applications*, *38*(2), 322–342. https://doi.org/10.1137/16M1074485

DRAFT